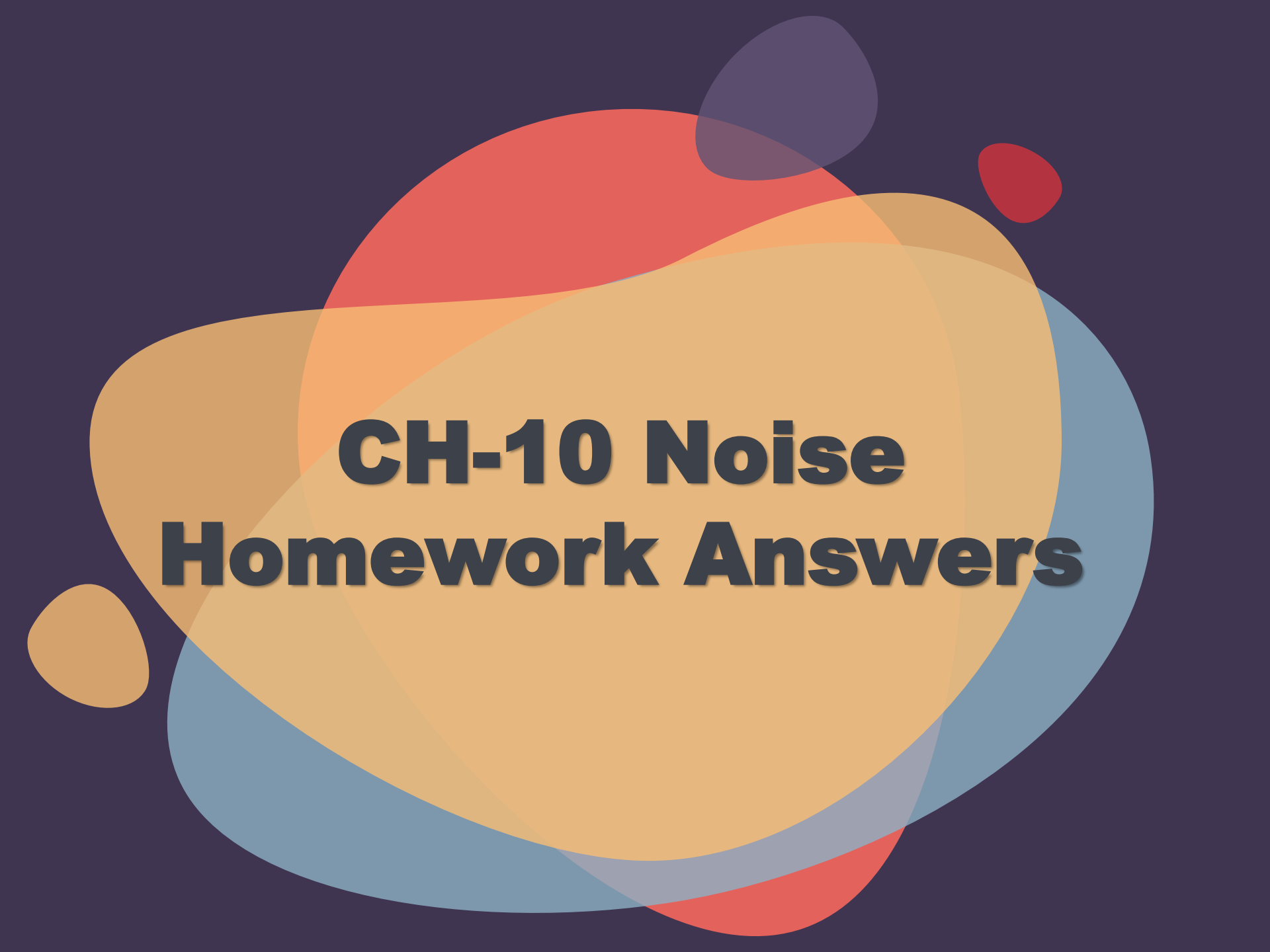




CH-10 Noise Homework Answers

01

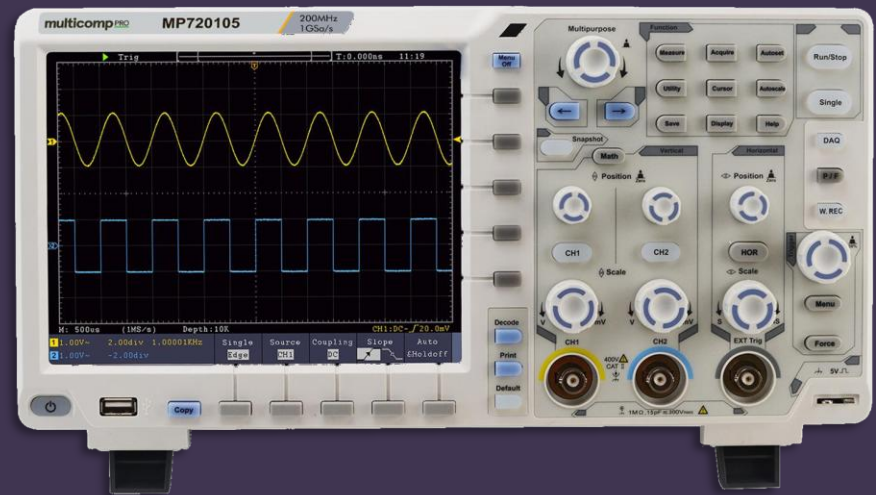
Of the following, which sound measuring instrument is the LEAST commonly used for most noise problems encountered in a manufacturing plant?

A. Sound level meter

B. Octave-band analyzer

C. Noise dosimeter

D. Oscilloscope





02

An octave band analyzer is an instrument used to determine:

A. The noise frequency distribution

B. SPL in A, B, and C weighted scales

C. Peak to peak SPL's

D. A sound level within a fraction of a decibel



03

The correct term for hearing loss that can be induced by occupational exposure to noise is:

A. Presbycusis

B. Sociocusis

C. Sensorineural

D. Conductive

04

The best method for monitoring noise exposure of a long-haul truck driver is through the use of

- A. Sound level meter
- B. Octave band analyzer
- C. Noise dosimeter**
- D. Annual Audiogram



05

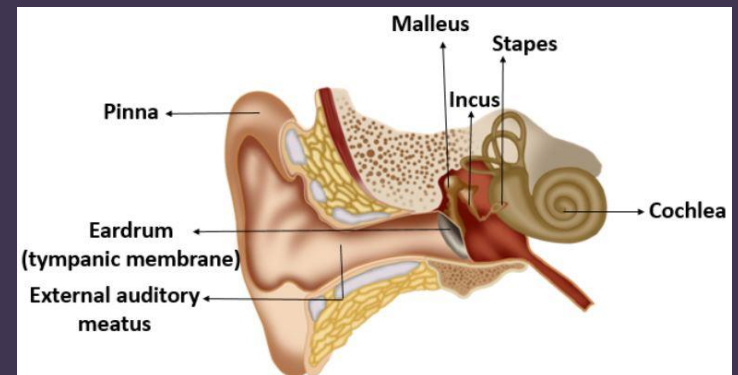
The small bones in the middle ear that transmit mechanical vibration from the eardrum to the cochlea are:

A. Incas, stapeziod, and mellous

B. Anvil, horseshoe, and wrench

C. Malleus, incus, and stapes

D. None of the above





06

The normal range of human hearing is:

A. 20 to 2,000 Hz

B. 20 to 20,000 Hz

C. 20 to 200,000 Hz

D. 200 to 200,000 Hz

07

In a work area the noise levels are 95 dBA for 2hr, 90 dBA for 4hr and 85 dBA for 2hr.

Does this exceed the OSHA limits?

$$D = 100 \left[\sum_{i=1}^N \frac{C_i}{T_i} \right]$$

A. Yes

B. No

C. Can't determine from information given

D. Borderline exposure

$$D = 100 \left(\frac{2}{4} + \frac{4}{8} + \frac{2}{16} \right)$$

$$D = 100 \left(\frac{8+8+2}{16} \right) = 100 \times \frac{18}{16}$$

$$D = 100 \times 1.125 = 112.5\%$$



08

What is the overall noise level to be expected when two machines are operating simultaneously in a room if the SPL due to machine X is 95 dB and the SPL due to machine Y is 99 dB?

$$L_{PT} = 10 \log \left(\sum_{i=1}^N 10^{\frac{L_{Pi}}{10}} \right)$$

A. 95 dB

B. 97 dB

C. 99 dB

D. 101 dB

$$L_{PT} = 10 \log \left(\sum_{i=1}^N \left(10^{\frac{95}{10}} + 10^{\frac{99}{10}} \right) \right)$$

$$L_{PT} = 10 \times 10.0455 = 100.455 \cong 101$$



09

The sound level from a fire alarm was measured at 115 dBA at a distance of 3 feet. What is the estimated sound level at a distance of 20 feet?

$$dB_1 = dB_0 + 20 \log_{10} \left(\frac{d_0}{d_1} \right)$$

$$dB_1 = dB_0 + 20 \log \left(\frac{d_0}{d_1} \right)$$

$$dB_1 = 115 + 20 \log \left(\frac{3}{20} \right)$$

$$dB_1 = 115 + 20 \log(0.15)$$

$$dB_1 = 115 + 20 x - 0.8239$$

$$dB_1 = 115 + (-16.478)$$

$$dB_1 = 98.52$$

A. 95.0 dBA

B. 98.5 dBA

C. 114 dBA

D. 135 dBA



10

A small business maintains a computer server to facilitate email and data storage. The fan on the server fan generates noise measured at 86 dBA at a nearby worker's desk. Which noise control solution should the business owner try first to lower the employee's noise exposure?

- A. Issue hearing protection devices with an NRR greater than 5 to the worker
- B. Buy a radio for the worker so that the employee can listen to music over the noise of the server
- C. Build an enclosure around the server to contain the noise
- D. Change the orientation of the server to direct the noise in a different direction**